

Diego Eduardo Martínez Cruz

Robotics & Automation Technician — Field Service Engineer

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PROFESSIONAL SUMMARY

Mechatronics professional specializing in industrial automation, robotics integration, and field service engineering. Hands-on command of the full control stack — PLC and VFD programming, robotic cells, electrical power, and on-site commissioning — built on a strong physics, mathematics, and CAD/simulation foundation. Proven ability to diagnose critical failures, lead installations, and bring equipment online under CFE and NOM/STPS standards, ensuring uptime and process efficiency.

WORK EXPERIENCE

Founder & Developer — Logic PLC (logicplc.vercel.app) 2025 – Present

- Built a web platform for AI-assisted PLC programming that generates and debugs industrial control logic from natural language.
- Led the product end to end: architecture, UX, and integration of AI models (LLMs) for automation workflows.

Automation & Field Service Engineer — Badger Automation, Kotkoff, Molinera Del Valle Feb 2025 – Dec 2025
Clients: Kellogg's, Bimbo, and Intuitive Surgical.

- Fault diagnosis (troubleshooting), signal tracing (I/O checks), and programming of PLCs and variable-frequency drives.
- Technical site surveys and full installation of industrial machinery, validating electrical, pneumatic, and hydraulic infrastructure.
- Correction of electrical control diagrams during commissioning; panel maintenance and electrical installation work under CFE standards.

Electromechanical Technician — Jeneza, Mexicali, B.C. Oct 2022 – Jan 2025
Clients: Jonathan Engineers, Chromalloy, and Bachoco.

- Preventive and corrective maintenance of electrical substations, transformers, and load centers in industrial environments.
- Installation, adjustment, and repair of key industrial equipment, including squirrel-cage motors.
- Applied strict safety standards (LOTO, high-risk work), coordinating contractors during shutdown windows.

KEY ENGINEERING SKILLS

- Automation & Controls:** PLC programming (Allen-Bradley, Studio 5000), VFDs, HMI development, PID control loops, I/O troubleshooting.
- Robotics:** Industrial robot integration (Fanuc, Kuka), robotic-cell design & kinematics simulation, ROS 2.
- Design & Simulation:** SolidWorks (CSWA/CSWP), AutoCAD & AutoCAD Electrical, Fusion 360, basic GD&T.
- Electrical & Electromechanical:** LV/MV systems, substations & transformers, motor control, grounding, embedded software.
- Standards & Languages:** CFE, NOM/STPS, LOTO. Spanish (Native), English (Intermediate Technical).

EDUCATION & CERTIFICATIONS

- Mechatronics Engineering** — Instituto Tecnológico de Mexicali (ITM) 2022 – Present
- Physics & Mathematics** — UABC, Ensenada 2020 – 2022
- Technical Certifications:** Software Engineer — HackerRank (2025); CSWA & CSWP — SolidWorks; PLC Programming — Allen-Bradley.
- Courses:** TIG Welding & Joining Processes (Universidad 16 de Septiembre); Biorobotic & 3D Systems Design (PH4 BIONICS).

OCCUPATIONAL SAFETY CERTIFICATIONS (DC-3, STPS)

- Electrical & Equipment Safety:** Live Electrical Work; Lockout/Tagout (LOTO, NOM-004-STPS-1999); Machinery, Equipment & Tools Safety (NOM-004-STPS); Lifting Platforms.
- High-Risk Work:** Working at Heights; Confined Spaces; Hot Work (Oxy-Cutting & Welding); Fire Prevention & Extinguishers.
- Hazmat & Others:** Chemical Hazard Communication (NOM-018-STPS-2015); Construction Safety (NOM-031-STPS-2011); Ergonomic Risk — Manual Handling (NOM-036-1-STPS-2018); Personal Protective Equipment (NOM-017-STPS-2008).

ENGINEERING PROJECTS

- Robotic Manufacturing Cell — Design & Simulation (2025):** Flexible spot-welding cell in SolidWorks and AutoCAD; robot kinematics simulation and cycle-time analysis.
- PID Process Control System (2024):** PID temperature control for a heat exchanger; loop programmed on PLC with an HMI for monitoring and setpoint tuning.
- Exploratory Rover Design (2024):** Rocker-Bogie suspension with motion design and analysis in Fusion 360. Advanced Robotics Club, ITM.